



GIRRAWEE HIGH SCHOOL

MATHEMATICS EXTENSION 2

TASK 1 2012 – COMPLEX NUMBERS

ANSWERS COVER SHEET

FINAL MARK

Name: _____

Teacher: _____

	MARK	E1	E2	E3	E4	E5	E6	E7	E8	E9
1	/7	✓		✓						✓
	/7									
2	/9	✓		✓						✓
	/9									
3	/18	✓		✓						✓
	/18									
4	/11	✓		✓						✓
	/11									
5	/16	✓		✓						✓
	/16									
6	/18	✓		✓						✓
	/18									
7	/13	✓		✓						✓
	/13									
8	/15	✓		✓						✓
	/15									
9	/9	✓		✓						✓
	/9									
TOTAL	/116	/116				/116				/116

- E1 appreciates the creativity, power and usefulness of mathematics to solve a broad range of problems.
- E2 chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings.
- E3 uses the relationship between algebraic and geometric representations of complex numbers and of conic sections.
- E4 uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections and polynomials.
- E5 uses ideas and techniques from calculus to solve problems in mechanics involving resolution of forces, resisted motion and circular motion
- E6 combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions.
- E7 uses the techniques of slicing and cylindrical shells to determine volumes.
- E8 applies further techniques of integration, including partial fractions, integration by parts and recurrence formulae, to problems.
- E9 communicates abstract ideas and relationships using appropriate notation and logical argument.

Girraween High School

Year 12

Mathematics Extension 2

Task 1 December 2011 – Complex Numbers

Instructions:

*Attempt all questions.

*Write all answers on your own paper. Remember to start each of Questions 1, 2, 3, 4, 5 and 6 on a separate sheet of paper.

*Show all necessary working.

*Marks may be deducted for careless or badly arranged work.

*Time allowed: 100 minutes

Question 1 (7 Marks)

Marks

Given that $z = 5 - 3i$ and $w = 4 + i$ find

- | | |
|-------------------|---|
| (a) $2z + 3w$ | 1 |
| (b) zw | 1 |
| (c) $w\bar{z}$ | 2 |
| (d) z^2 | 1 |
| (e) $\frac{z}{w}$ | 2 |

Question 2 (9 Marks)

Given that $z = i\sqrt{3} - 1$ and $w = 2 + 2i$

- | | |
|--|---|
| (a) Find zw in Cartesian ($x + iy$) form | 1 |
| (b) Convert z and w to modulus/ argument form. | 4 |
| (c) Hence find zw in modulus/argument form and use your result to find | 4 |

the exact value of $\cos \frac{11\pi}{12}$.

Question 3 (18 Marks)**Marks**

- (a) (i) Convert $-2 + 2i\sqrt{3}$ to modulus/ argument form. 2
- (ii) Hence find the value of $(-2 + 2i\sqrt{3})^7$ in Cartesian form. 4
- (b) (i) Find all real numbers x and y such that $(x + iy)^2 = 3 - 4i$. 5
- (ii) Hence solve the quadratic equation $z^2 + (4 + 3i)z + (1 + 7i) = 0$ 3
- (c) Find the five 5th roots of $16\sqrt{3} - 16i$. Leave your answers in 4
- modulus/ argument form.

Question 4 (11 Marks)

Sketch these regions on separate Argand diagrams:

- (a) $|z - (3 + 4i)| < 5$ 3
- (b) $-\frac{\pi}{6} < \text{Arg}(z - 2) \leq \frac{\pi}{2}$ 4
- (c) $2 \leq |z| < 3$ and $\text{Arg} z < -\frac{\pi}{2}$ 4

Question (5) (16 Marks)

(a) Find the Cartesian equations for the following loci:

- (i) $|z - 2i + 3| = 6$ 2
- (ii) $|z + 4i| = |z - 3|$ 3

Question 5 continues on the next page

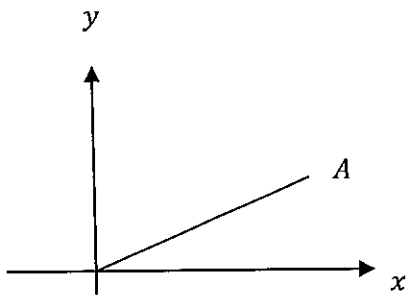
Question 5 (continued)

Marks

(b) z is a complex number so that $\overrightarrow{OA} = z$. Copy the Argand diagram

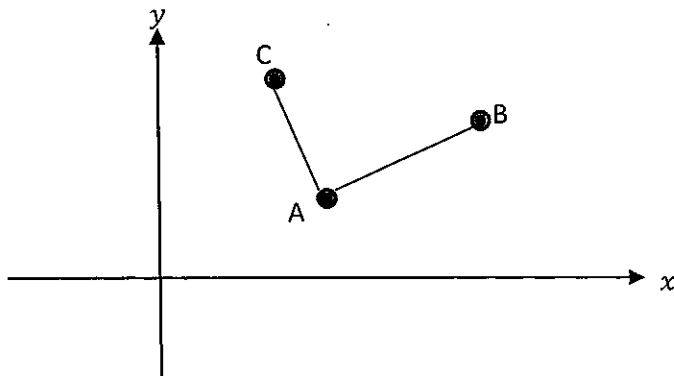
below on to your working paper and draw in

- | | | |
|-------|-------------------------------|---|
| (i) | $\overrightarrow{OB} = z + 1$ | 1 |
| (ii) | $\overrightarrow{OC} = z - i$ | 1 |
| (iii) | $\overrightarrow{OD} = iz$ | 1 |
| (iv) | $\overrightarrow{OE} = -z$ | 1 |



(c) A, B, C and D are points on an Argand diagram such that $AB = AC$

and $\angle BAC = \frac{\pi}{2}$. (See diagram below). If $\overrightarrow{OA} = z_1$ and $\overrightarrow{OB} = z_2$,



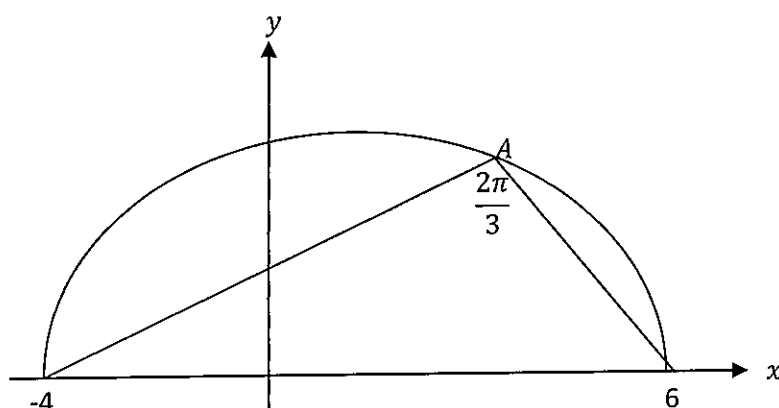
- | | | |
|-------|---|---|
| (i) | Find an expression for $\overrightarrow{OC}$ in terms of z_1 and z_2 . | 1 |
| (ii) | Find an expression for $\overrightarrow{OD}$ in terms of z_1 and z_2 if $ABDC$ is a square. | 1 |
| (iii) | If $ABCD$ is a square show that $AD \times BC = 2 z_2 - z_1 ^2$ and that $\frac{\overrightarrow{BC}}{\overrightarrow{AD}}$ is entirely imaginary. | 5 |

Question 6 (18 Marks)

Marks

- (a) The locus $\text{Arg} \left(\frac{z-6}{z+4} \right) = \frac{2\pi}{3}$ represents part of a circle (see below.) 5

If $\overrightarrow{OA} = z$ find the centre and radius of the circle and the Cartesian equation for the locus of z . (Note that this is NOT a semicircle.)



- (b) If $z = \cos \theta + i \sin \theta$
- (i) Show that $z - \frac{1}{z} = 2i \sin \theta$. 2
 - (ii) Show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$ 2
 - (iii) Hence express $\sin^6 \theta$ in terms of $\cos 6\theta$, $\cos 4\theta$ and $\cos 2\theta$. 4
- (c) If $w = x + iy$ and $w = \frac{z-2i}{1-z}$
- (i) Show that $z = \frac{x+i(y+2)}{(x+1)+iy}$ 2
 - (ii) Find the locus of w if $|z| = 2$. 3

Question 7 (13 Marks)

- (a) If w is the non real root of $z^7 - 1 = 0$ with the smallest possible positive argument:
- (i) Show that $w = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$ and that w^2, w^3, w^4, w^5 and w^6 are the other non real roots of $z^7 - 1 = 0$. 3
 - (ii) Show that $w + w^2 + w^3 + w^4 + w^5 + w^6 = -1$ 1
 - (iii) By expanding $(w + w^6)(w^2 + w^5)(w^3 + w^4)$ show that $(w + w^6)(w^2 + w^5)(w^3 + w^4) = 1$. 4
- Hence show that $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{3\pi}{7} = \frac{1}{8}$.
- (b) Resolve $z^8 + 1$ into real quadratic factors. Hence show that 5
- $$\cos 4\theta = 8(\cos \theta - \cos \frac{\pi}{8})(\cos \theta + \cos \frac{\pi}{8})(\cos \theta - \cos \frac{3\pi}{8})(\cos \theta + \cos \frac{3\pi}{8})$$

(You do NOT need to prove $z^n + \frac{1}{z^n} = 2 \cos n\theta$ if $|z| = 1$ this time.)

Question 8 (15 Marks)

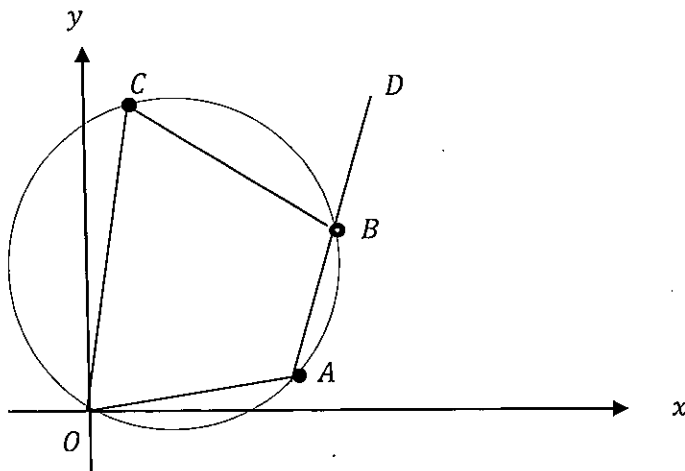
Marks

(a) (i) By expanding $(\cos\theta + i\sin\theta)^5$ show that $\tan 5\theta = \frac{5\tan\theta - 10\tan^3\theta + \tan^5\theta}{1 - 10\tan^2\theta + 5\tan^4\theta}$ 3

(ii) Hence show that $\tan \frac{\pi}{5}$ is a solution of $z^4 - 10z^2 + 5 = 0$ 3

(iii) Hence find the exact value of $\tan \frac{\pi}{5}$. 4

- (b) z_1, z_2 and z_3 are represented by the vectors $\overrightarrow{OA}, \overrightarrow{OB}$ and $\overrightarrow{OC}$ respectively.
If O, A, B and C are the vertices of a circle and AB is extended to D (see below)



- (i) Show that $\text{Arg} \left(\frac{z_3}{z_1} \right) = \text{Arg} \left(\frac{z_3 - z_2}{z_2 - z_1} \right)$ 3
- (ii) Show that if $\overrightarrow{OE} = \frac{1}{z_1}, \overrightarrow{OF} = \frac{1}{z_2}, \overrightarrow{OG} = \frac{1}{z_3}$, then E, F , and G are in a straight line. 2

Question 9 (9 Marks)

- (i) Show that $\frac{1 + \cos\theta + i\sin\theta}{1 - \cos\theta - i\sin\theta} = \frac{i\sin\theta}{1 - \cos\theta}$. 2
- (ii) Hence or otherwise show that $\frac{1 + \cos\theta + i\sin\theta}{1 - \cos\theta - i\sin\theta} = i \cot \left(\frac{\theta}{2} \right)$ 2
- (iii) Hence or otherwise show that if $z = i \cot \left(\frac{\theta}{2} \right)$ then $\frac{z-1}{z+1} = \cos\theta + i\sin\theta$. 2
- (iv) Hence solve the equation $\left(\frac{z-1}{z+1} \right)^4 = -1$. 3

END OF EXAMINATION

Q. (1) (a) $2z + 3w$

$= 2(5-i) + 3(4+i)$

$= 10 - 2i + 12 + 3i$

$= 22 - 3i$

Q. (2) (a)

$z = w$

$= (-1 + i\sqrt{3})(2 + i)$

$= -2 - 2i + 2i\sqrt{3} - 2\sqrt{3}$

$= -2 - 2\sqrt{3} + i(2\sqrt{3} - 2)$

(b) zw

$= (5-i)(4+i)$

$= 20 + 5i - 12i + 3$

$= 23 - 7i$

(c) wz

$= (4+i)(5+i)$

$= 20 + 12i + 5i + 3$

$= 17 + 17i$ or $17(1+i)$

(d) z^2

$= (5-i)^2$

$= 25 - 30i + 9$

$= 16 - 30i$

(e) $\frac{z}{w}$

$= \frac{(5-i)}{(4+i)} \times \frac{4-i}{4-i}$

$= \frac{20 - 5i - 12i + 3}{17}$

$= \frac{23 - 17i}{17}$

$= \frac{23}{17} - i$

$= \frac{23}{17} - i$

$= 1 - i$

(b) $|z| = \sqrt{3^2 + (\sqrt{3})^2}$

$= 2$

Arg $z = \tan^{-1}(\frac{\sqrt{3}}{3})$ [But Q. 2]

$= \frac{2\pi}{3}$

$z = 2 \operatorname{cis} \frac{2\pi}{3}$

$|w| = \sqrt{2^2 + 2^2}$

$= 2\sqrt{2}$

Arg $w = \tan^{-1}(\frac{2}{2})$

$= \frac{\pi}{4}$

$w = 2\sqrt{2} \operatorname{cis} \frac{\pi}{4}$

(c) z^2

$= 2 \operatorname{cis} \frac{2\pi}{3} \times 2 \operatorname{cis} \frac{2\pi}{3}$

$= 4 \operatorname{cis} (\frac{4\pi}{3} + i \sin \frac{4\pi}{3})$

(2) (c) (continued)

Equating real parts of answers

$4\sqrt{2} \cos \frac{4\pi}{3} = -2 - 2\sqrt{3}$

$\therefore \cos \frac{4\pi}{3} = \frac{-2 - 2\sqrt{3}}{4\sqrt{2}}$

$\cos \frac{4\pi}{3} = \frac{(-2 - 2\sqrt{3})}{4\sqrt{2}}$

$\cos \frac{4\pi}{3} = \frac{(-2 - 2\sqrt{3})}{4\sqrt{2}}$

(3) (a) (i) $-2 + 2i\sqrt{3}$

$= 4 \operatorname{cis} \frac{2\pi}{3}$

(ii) By De Moivre's Theorem

$(-2 + 2i\sqrt{3})^3$

$= (4 \operatorname{cis} \frac{2\pi}{3})^3$

$= 4^3 \operatorname{cis} \frac{14\pi}{3}$

$= 16 \operatorname{cis} (\frac{14\pi}{3} - 2\pi)$

$= -8192 + 8192i\sqrt{3}$

$$Q. (3)(i) (2+iy)^2 = 3-4i$$

$$\therefore (x^2 - y^2) + 2ixy = 3 - 4i$$

$$x^2 - y^2 = 3 \quad (1) \text{ Equating reals}$$

$$2xy = -4 \quad (2) \text{ Equating imaginaries}$$

$$\text{Using (2)}: y = -\frac{2}{x}$$

$$\text{Sub (2) in (1)}: x^2 - \left(-\frac{2}{x}\right)^2 = 3$$

$$x^2 - \frac{4}{x^2} = 3$$

$$x^2 \times 6x^2 \text{ by } x^2$$

$$x^4 - 4 = 3x^2$$

$$x^4 - 3x^2 - 4 = 0$$

$$(x^2 - 4)(x^2 + 1) = 0$$

$$\text{As } x \text{ is real } x = \pm 2$$

$$\text{As } y = -\frac{2}{x}, y = \mp 1$$

$$\therefore \text{The numbers are } x = 2, y = -1$$

$$\& x = -2, y = 1$$

$$\therefore \text{The 2 square roots of } 3-4i \text{ are}$$

$$2-i \& -2+i$$

(ii) Hence as the solutions to

$$z^2 + (4+3i)z + (1+7i) = 0 \text{ are}$$

$$z = \frac{-(4+3i) \pm \sqrt{(4+3i)^2 - 4 \times 1 \times (1+7i)}}{2}$$

$$= \frac{-4-3i \pm \sqrt{16+24i-9-4-28i}}{2}$$

$$= \frac{-4-3i \pm \sqrt{3-4i}}{2}$$

$$= \frac{-4-3i \pm (2-i)}{2} \quad [\text{from (i)}]$$

$$= \frac{-4-3i+2-i}{2} \text{ or } \frac{-4-3i-2+i}{2} = \frac{-2-4i}{2} \text{ or } \frac{-6-2i}{2}$$

$$z = -1-2i \text{ or } z = -3-i$$

$$Q. (3)(i) 16\sqrt{3} - 16i = 32\left(\cos -\frac{\pi}{6} + i \sin -\frac{\pi}{6}\right)$$

$$\therefore \text{The 5th roots are } 2\left(\cos \frac{-\pi}{30} + i \sin \frac{-\pi}{30}\right)$$

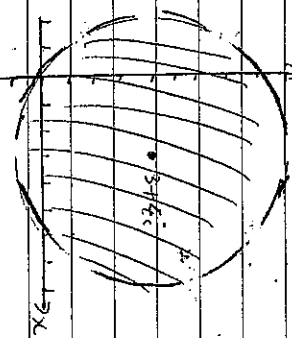
$$2\left(\cos \frac{11\pi}{30} + i \sin \frac{11\pi}{30}\right)$$

$$2\left(\cos \frac{23\pi}{30} + i \sin \frac{23\pi}{30}\right)$$

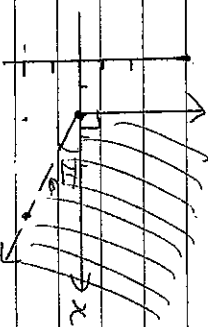
$$2\left(\cos \frac{35\pi}{30} + i \sin \frac{35\pi}{30}\right)$$

$$2\left(\cos \frac{47\pi}{30} + i \sin \frac{47\pi}{30}\right)$$

$$Q. (4)(a) |z - (3+4i)| < 5$$

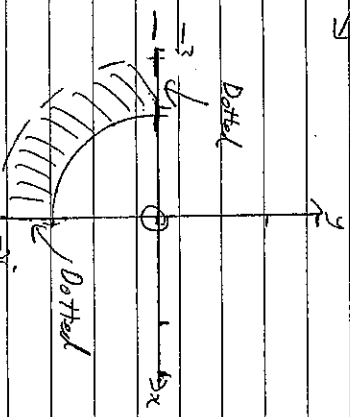


$$(b) -\frac{\pi}{6} < \text{Arg}(z-2) \leq \frac{\pi}{2}$$

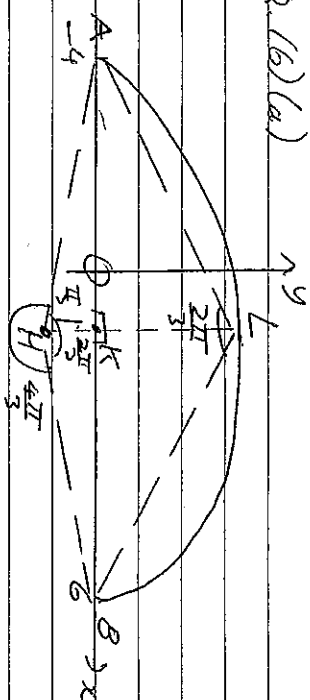


$$(c) 2 \leq |z| < 3$$

$$\text{and } \text{Arg } z < -\frac{\pi}{2}$$



Q.16)(a)



Mod

Note: Centre of circle will be on line $x=1$

(By symmetry)

If H is circle centre, $A = -4$ & $B = 6$

then $\angle AHB [\text{reflex}] = \frac{4\pi}{3}$ [at centre $= 2x$]

Let circumference

on same arc

$$\therefore \angle AHB [\text{obtuse}] = \frac{2\pi}{3} [\text{on same arc}]$$

Letting $1 = K_3$

& point at $x = \text{lower edge of circle} = 1$

$\angle AHK = \frac{\pi}{3}$ (By symmetry)

$$\therefore \angle KAH = \frac{\pi}{6} [\text{in } \triangle KAH]$$

$$\therefore KH = 5 \tan \frac{\pi}{6}$$

$$= \frac{5}{\sqrt{3}}$$

$$\therefore KA = \sqrt{5^2 + \left(\frac{5}{\sqrt{3}}\right)^2}$$

$$= \frac{10\sqrt{3}}{3}$$

$$\therefore \text{Locus of } z = (x-1)^2 + \left(y + \frac{5\sqrt{3}}{3}\right)^2 = \frac{100}{3}, y > 0$$

Q.16)(b) (i) $z - \frac{1}{z}$

$$= (\cos \theta + i \sin \theta) - (\cos \theta + i \sin \theta) \quad [\text{By De Moivre}]$$

$$= (\cos \theta + i \sin \theta) - (\cos \theta - i \sin \theta) \quad [\text{as } \cos \theta \text{ even}]$$

$$= 2i \sin \theta$$

$$(ii) \quad z^n + \frac{1}{z^n}$$

$$= (\cos \theta + i \sin \theta)^n + (\cos \theta - i \sin \theta)^n \quad [\text{By De Moivre}]$$

$$= (\cos^n \theta + i \sin^n \theta) + (\cos^n \theta - i \sin^n \theta) \quad [\text{as } \cos \theta \text{ even}]$$

$$= 2 \cos^n \theta$$

$$(iii) \text{ If } z = \cos \theta + i \sin \theta$$

$$\left(z - \frac{1}{z}\right)^6$$

$$= (2i \sin \theta)^6 \quad [\text{using (i)}]$$

$$= -64 \sin^6 \theta \quad (1)$$

$$\text{As } \left(z - \frac{1}{z}\right)^6$$

$$= z^6 - 6z^4 + 15z^2 - 20 + \frac{15}{z^2} - \frac{6}{z^4} + \frac{1}{z^6}$$

$$= \left(z^6 + \frac{1}{z^6}\right) - 6\left(z^4 + \frac{1}{z^4}\right) + 15\left(z^2 + \frac{1}{z^2}\right) - 20$$

$$= 2 \cos 6\theta - 12 \cos 4\theta + 30 \cos 2\theta - 20 \quad [\text{using (ii)}] \quad (2)$$

Equating (1) & (2)

$$-64 \sin^6 \theta = 2 \cos 6\theta - 12 \cos 4\theta + 30 \cos 2\theta - 20$$

$$\therefore \sin^6 \theta = \frac{-\cos 6\theta + 3 \cos 4\theta - 15 \cos 2\theta + 5}{16}$$

$$Q. (6) (i) w = \frac{z-2i}{1-z}$$

$$\therefore w - wz = z - 2i$$

$$w + 2i = z + wz$$

$$w + 2i = z(1+w)$$

$$\frac{w+2i}{1+w} = z$$

$$\text{As } w = x + iy$$

$$z = \frac{x+iy+2i}{(x+1)+iy}$$

$$(x+1)+iy$$

$$(ii) \text{ If } |z| = 2 \text{ then } \left| \frac{x+iy+2i}{(x+1)+iy} \right| = 2$$

Squaring B.S

$$\frac{x^2 + (y+2)^2}{(x+1)^2 + y^2} = 4$$

$$x^2 + y^2 + 4y + 4 = 4x^2 + 8x + 4 + 4y^2$$

$$0 = 3x^2 + 8x + 3y^2 - 4y$$

$$0 = x^2 + \frac{8}{3}x + y^2 - \frac{4}{3}y$$

$$\frac{20}{9} = (x^2 + \frac{8}{3}x + \frac{16}{9}) + (y^2 - \frac{4}{3}y + \frac{4}{9})$$

$\therefore$ locus of w is

$$(x + \frac{4}{3})^2 + (y - \frac{2}{3})^2 = \frac{20}{9}$$

$$Q. (7) (a) (i) \text{ If } w \text{ is a root of } z^7 - 1 = 0$$

$$\text{Roots} = \cos 0, \cos \frac{2\pi}{7}, \cos \frac{4\pi}{7}, \cos \frac{6\pi}{7}, \cos \frac{8\pi}{7}, \cos \frac{10\pi}{7}, \cos \frac{12\pi}{7}$$

Clearly, $\cos \frac{2\pi}{7}, \cos \frac{4\pi}{7}, \cos \frac{6\pi}{7}$ is non real root with smallest possible positive argument

$$w^2 = (\cos \frac{2\pi}{7})^2 = \cos \frac{4\pi}{7} \text{ which is a root of } z^7 - 1 = 0$$

$$w^3 = (\cos \frac{2\pi}{7})^3 = \cos \frac{6\pi}{7} \quad " \quad " \quad "$$

$$w^4 = (\cos \frac{2\pi}{7})^4 = \cos \frac{8\pi}{7} \quad " \quad " \quad "$$

$$w^5 = (\cos \frac{2\pi}{7})^5 = \cos \frac{10\pi}{7} \quad " \quad " \quad "$$

$$w^6 = (\cos \frac{2\pi}{7})^6 = \cos \frac{12\pi}{7} \quad " \quad " \quad "$$

Hence w^2, w^3, w^4, w^5, w^6 are the other non real roots.

(ii) By sum of roots of $z^7 - 1 = 0$

$$[x^6 + \dots = -b]$$

$$1 + w + w^2 + w^3 + w^4 + w^5 + w^6 = 0$$

$$\text{Hence } w + w^2 + w^3 + w^4 + w^5 + w^6 = -1$$

$$(iii) (w + w^6)(w^2 + w^5)(w^3 + w^4)$$

$$= (w + w^6)(w^8 + w^{11})(w^3 + w^4)$$

$$= w^6 + w + w^9 + w^{10} + w^{11} + w^{12} + w^{14} + w^{15}$$

$$= w^6 + 1 + w^3 + w^3 + w^4 + w^5 + 1 + w$$

$$= 1 + (1 + w + w^2 + w^3 + w^4 + w^5 + w^6)$$

$$= 1 \quad \text{As } 1 + w + w^2 + w^3 + w^4 + w^5 + w^6 = 0 \text{ from (i)} \quad (1)$$

Also as $w + w^6 = 2 \cos \frac{2\pi}{7}$

$$= (\cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} + \cos \frac{12\pi}{7} + i \sin \frac{12\pi}{7})$$

$$= (\cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} + \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7})$$

$$= 2 \cos \frac{2\pi}{7} \quad (2)$$

Similarly, $w^2 + w^5 = 2 \cos \frac{4\pi}{7}$

$$[= -2 \cos \frac{3\pi}{7}] \quad (3)$$

$$\& w^3 + w^4 = 2 \cos \frac{6\pi}{7}$$

$$[= -2 \cos \frac{\pi}{7}] (4) \text{ P.T.O } \rightarrow$$

Q. (7) (a) (i) [continued]

As $(w + w^6)(w^2 + w^5)(w^3 + w^4) = 1$

$2 \cos \frac{2\pi}{7} x - 2 \cos \frac{3\pi}{7} x - 2 \cos \frac{4\pi}{7} x = 1$

$8 \cos \frac{2\pi}{7} \cos \frac{3\pi}{7} \cos \frac{4\pi}{7} = 1$

$\cos \frac{2\pi}{7} \cos \frac{3\pi}{7} \cos \frac{4\pi}{7} = \frac{1}{8}$

(7) (b) Roots of $z^8 + 1 = 0$ are

$z = \cos \frac{7\pi}{8} + i \sin \frac{7\pi}{8}, k = 0, 1, 2, \dots, 7$

$z^8 + 1 = (z - \cos \frac{7\pi}{8} - i \sin \frac{7\pi}{8})(z - \cos \frac{9\pi}{8} - i \sin \frac{9\pi}{8})(z - \cos \frac{11\pi}{8} - i \sin \frac{11\pi}{8})(z - \cos \frac{13\pi}{8} - i \sin \frac{13\pi}{8})(z - \cos \frac{15\pi}{8} - i \sin \frac{15\pi}{8})(z - \cos \frac{17\pi}{8} - i \sin \frac{17\pi}{8})(z - \cos \frac{19\pi}{8} - i \sin \frac{19\pi}{8})(z - \cos \frac{21\pi}{8} - i \sin \frac{21\pi}{8})$

$(z^2 - 2 \cos \frac{7\pi}{8} z + 1)(z^2 - 2 \cos \frac{9\pi}{8} z + 1)(z^2 - 2 \cos \frac{11\pi}{8} z + 1)(z^2 - 2 \cos \frac{13\pi}{8} z + 1)$

$(z^2 - 2 \cos \frac{7\pi}{8} z + 1) \quad (1)$

Note: As $(z - \cos \frac{7\pi}{8} - i \sin \frac{7\pi}{8})(z - \cos \frac{15\pi}{8} - i \sin \frac{15\pi}{8}) = z^2 - 2 \cos \frac{7\pi}{8} z + 1 + i \sin \frac{7\pi}{8} - i \sin \frac{15\pi}{8} = z^2 - 2 \cos \frac{7\pi}{8} z + 1$

$z^2 - 2 \cos \frac{7\pi}{8} z + 1$

Similarly $(z - \cos \frac{3\pi}{8} - i \sin \frac{3\pi}{8})(z - \cos \frac{13\pi}{8} - i \sin \frac{13\pi}{8}) = z^2 - 2 \cos \frac{3\pi}{8} z + 1$

$(z - \cos \frac{5\pi}{8} - i \sin \frac{5\pi}{8})(z - \cos \frac{11\pi}{8} - i \sin \frac{11\pi}{8}) = z^2 - 2 \cos \frac{5\pi}{8} z + 1$

$(z - \cos \frac{7\pi}{8} - i \sin \frac{7\pi}{8})(z - \cos \frac{9\pi}{8} - i \sin \frac{9\pi}{8}) = z^2 - 2 \cos \frac{7\pi}{8} z + 1$

Continuing with (1) $\cos \frac{5\pi}{8} = -\cos \frac{3\pi}{8}$ & $\cos \frac{7\pi}{8} = -\cos \frac{3\pi}{8}$

$z^8 + 1 = (z^2 - 2 \cos \frac{7\pi}{8} z + 1)(z^2 - 2 \cos \frac{9\pi}{8} z + 1)(z^2 - 2 \cos \frac{11\pi}{8} z + 1)(z^2 - 2 \cos \frac{13\pi}{8} z + 1)(z^2 - 2 \cos \frac{15\pi}{8} z + 1)(z^2 - 2 \cos \frac{17\pi}{8} z + 1)(z^2 - 2 \cos \frac{19\pi}{8} z + 1)(z^2 - 2 \cos \frac{21\pi}{8} z + 1)$

$z^8 + 1 = (z^2 - 2 \cos \frac{7\pi}{8} z + 1)(z^2 - 2 \cos \frac{9\pi}{8} z + 1)(z^2 - 2 \cos \frac{11\pi}{8} z + 1)(z^2 - 2 \cos \frac{13\pi}{8} z + 1)(z^2 - 2 \cos \frac{15\pi}{8} z + 1)(z^2 - 2 \cos \frac{17\pi}{8} z + 1)(z^2 - 2 \cos \frac{19\pi}{8} z + 1)(z^2 - 2 \cos \frac{21\pi}{8} z + 1)$

$z^4 + \frac{1}{z^4} = (z + \frac{1}{z} - 2 \cos \frac{7\pi}{8})(z + \frac{1}{z} - 2 \cos \frac{9\pi}{8})(z + \frac{1}{z} - 2 \cos \frac{11\pi}{8})(z + \frac{1}{z} - 2 \cos \frac{13\pi}{8})(z + \frac{1}{z} - 2 \cos \frac{15\pi}{8})(z + \frac{1}{z} - 2 \cos \frac{17\pi}{8})(z + \frac{1}{z} - 2 \cos \frac{19\pi}{8})(z + \frac{1}{z} - 2 \cos \frac{21\pi}{8})$

If $z = \cos \theta + i \sin \theta$, $z + \frac{1}{z} = 2 \cos \theta$ & $z + \frac{1}{z} = 2 \cos \theta$

Hence $2 \cos 4\theta = (2 \cos \theta - 2 \cos \frac{7\pi}{8})(2 \cos \theta - 2 \cos \frac{9\pi}{8})(2 \cos \theta - 2 \cos \frac{11\pi}{8})(2 \cos \theta - 2 \cos \frac{13\pi}{8})(2 \cos \theta - 2 \cos \frac{15\pi}{8})(2 \cos \theta - 2 \cos \frac{17\pi}{8})(2 \cos \theta - 2 \cos \frac{19\pi}{8})(2 \cos \theta - 2 \cos \frac{21\pi}{8})$

$(2 \cos \theta + 2 \cos \frac{7\pi}{8})(2 \cos \theta + 2 \cos \frac{9\pi}{8})(2 \cos \theta + 2 \cos \frac{11\pi}{8})(2 \cos \theta + 2 \cos \frac{13\pi}{8})(2 \cos \theta + 2 \cos \frac{15\pi}{8})(2 \cos \theta + 2 \cos \frac{17\pi}{8})(2 \cos \theta + 2 \cos \frac{19\pi}{8})(2 \cos \theta + 2 \cos \frac{21\pi}{8})$

$\cos 4\theta = 8 (\cos \theta - \cos \frac{7\pi}{8})(\cos \theta - \cos \frac{9\pi}{8})(\cos \theta - \cos \frac{11\pi}{8})(\cos \theta - \cos \frac{13\pi}{8})(\cos \theta - \cos \frac{15\pi}{8})(\cos \theta - \cos \frac{17\pi}{8})(\cos \theta - \cos \frac{19\pi}{8})(\cos \theta - \cos \frac{21\pi}{8})$

Q. (8) (a) (i)

$(\cos \theta + i \sin \theta)^5 = \cos^5 \theta + 5 \cos^4 \theta i \sin \theta - 10 \cos^3 \theta \sin^2 \theta + 10 \cos^2 \theta \sin^3 \theta - 5 \cos \theta \sin^4 \theta + i \sin^5 \theta$

$+ 5 \cos^4 \theta i \sin \theta + i \sin^5 \theta$

$\cos 5\theta + i \sin 5\theta = \cos^5 \theta + 5 \cos^4 \theta i \sin \theta - 10 \cos^3 \theta \sin^2 \theta + 10 \cos^2 \theta \sin^3 \theta - 5 \cos \theta \sin^4 \theta + i \sin^5 \theta$

$+ 5 \cos^4 \theta i \sin \theta + i \sin^5 \theta$ [By De Moivre]

$\frac{\sin 5\theta}{\cos 5\theta} = \frac{\sin 5\theta}{\cos 5\theta}$

$\tan 5\theta = \frac{\sin 5\theta}{\cos 5\theta} = \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta}$

$\tan 5\theta = \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta}$

$\tan 5\theta = \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta}$

(8) (a) (ii) Hence as $\tan \frac{\pi}{5} = \tan (5 \times \frac{\pi}{25}) = 0$

$\tan \frac{\pi}{5}$ is a solution to $\tan 5\theta = 0$

i.e. $5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta = 0$

$1 - 10 \tan^2 \theta + 5 \tan^4 \theta = 0$ by $1 - 10 \tan^2 \theta + 5 \tan^4 \theta$

$\tan^5 \theta - 10 \tan^3 \theta + 5 \tan \theta = 0$

$\tan \theta (\tan^4 \theta - 10 \tan^2 \theta + 5) = 0$

As $\tan \frac{\pi}{5} \neq 0$, $\tan \frac{\pi}{5}$ is a solution of $\tan^4 \theta - 10 \tan^2 \theta + 5 = 0$

Letting $z = \tan^2 \theta$, $z^2 - 10z + 5 = 0$

Hence solving $z^2 - 10z + 5 = 0$ (as a quadratic equation in z)

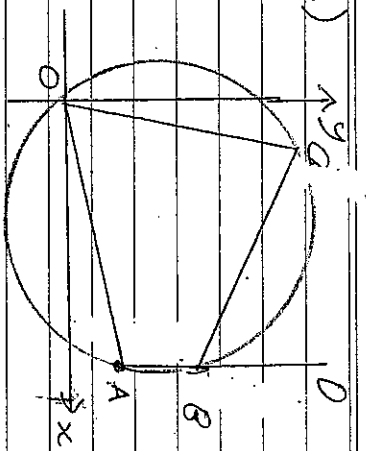
$z = \frac{10 \pm \sqrt{10^2 - 4 \times 1 \times 5}}{2 \times 1}$

$z = \frac{10 \pm \sqrt{80}}{2}$

$z^2 = 5 \pm 2\sqrt{5}$ or $z^2 = 5 - 2\sqrt{5}$

As $\frac{\pi}{5} < \frac{\pi}{2}$, $0 < \tan \frac{\pi}{5} < 1$

$\tan \frac{\pi}{5} = \sqrt{5 - 2\sqrt{5}}$



Q. (8) (b)

$$\angle AOC = \angle ABC$$

$$\angle AOC = \arg\left(\frac{z_3}{z_1}\right)$$

$$\angle CBD = \arg\left(\frac{z_3 - z_2}{z_2 - z_1}\right)$$

$\angle AOC = \angle CBD$ [exterior $\angle$ of cyclic quadrilateral]

$$\arg\left(\frac{z_3}{z_1}\right) = \arg\left(\frac{z_3 - z_2}{z_2 - z_1}\right)$$

(ii) If $\frac{1}{z_1}, \frac{1}{z_2}, \frac{1}{z_3}$ are in a straight line then

$$\arg\left(\frac{1}{z_3} - \frac{1}{z_2}\right) = \arg\left(\frac{1}{z_2} - \frac{1}{z_1}\right)$$

To prove this,

We know $\arg\left(\frac{z_3}{z_1}\right) = \arg\left(\frac{z_3 - z_2}{z_2 - z_1}\right)$

$$\arg\left(\frac{z_3}{z_1}\right) \times \frac{z_2 - z_1}{z_2 - z_1} = \arg\left(\frac{z_3 - z_2}{z_2 - z_1}\right) \times \frac{(z_2 - z_1)}{(z_2 - z_1)}$$

$$\arg\left(\frac{z_2 - z_1}{z_1 z_2}\right) = \arg\left(\frac{z_2 - z_3}{z_2 z_3}\right)$$

$$\arg\left(\frac{1}{z_1} - \frac{1}{z_2}\right) = \arg\left(\frac{1}{z_2} - \frac{1}{z_3}\right)$$

$\frac{1}{z_1}, \frac{1}{z_2}, \frac{1}{z_3}$ are in the same straight line

Q. (9) (i) $1 + \cos \theta + i \sin \theta \times (1 - \cos \theta + i \sin \theta)$
 $1 - \cos^2 \theta - i \sin^2 \theta \times (1 - \cos \theta + i \sin \theta)$

$$= (1 - \cos^2 \theta) + i \sin \theta + i \sin \theta \cos \theta + i^2 \sin^2 \theta - \sin^2 \theta$$

$$(1 - \cos^2 \theta)^2 + \sin^2 \theta$$

$$= \sin^2 \theta + 2i \sin \theta \cos \theta - \sin^2 \theta$$

$$1 - 2 \cos \theta + \cos^2 \theta + \sin^2 \theta$$

$$= \frac{2i \sin \theta}{2 - 2 \cos \theta}$$

$$= \frac{i \sin \theta}{1 - \cos \theta}$$

(ii) Using $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$
 $\& \cos \theta = 1 - 2 \sin^2 \frac{\theta}{2}$

$$\frac{i \sin \theta}{1 - \cos \theta}$$

$$= \frac{2i \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{1 - (1 - 2 \sin^2 \frac{\theta}{2})}$$

$$= \frac{2i \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \sin^2 \frac{\theta}{2}}$$

$$= \frac{i \cos \frac{\theta}{2}}{\sin \frac{\theta}{2}}$$

$$= i \cot \frac{\theta}{2}$$

PTD $\rightarrow$

Alternative (9)(i).

Otherwise for (9)(i)

$$\frac{1 + \cos \theta + i \sin \theta}{1 - \cos \theta - i \sin \theta}$$

As $\cos \theta = 2 \cos^2(\frac{\theta}{2}) - 1$

$\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$

$\therefore \cos \theta = 1 - 2 \sin^2(\frac{\theta}{2})$

$$= \frac{1 + 2 \cos^2(\frac{\theta}{2}) - 1 + 2 i \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{1 - (1 - 2 \sin^2 \frac{\theta}{2}) - 2 i \sin \frac{\theta}{2} \cos \frac{\theta}{2}}$$

$$= \frac{2 \cos^2(\frac{\theta}{2}) + 2 i \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \sin^2(\frac{\theta}{2}) - 2 i \sin \frac{\theta}{2} \cos \frac{\theta}{2}}$$

$$= \frac{2 \cos \frac{\theta}{2} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})}{2 \sin \frac{\theta}{2} (\sin \frac{\theta}{2} - i \cos \frac{\theta}{2})} \times \frac{\sin \frac{\theta}{2} + i \cos \frac{\theta}{2}}{\sin \frac{\theta}{2} + i \cos \frac{\theta}{2}}$$

$$= \cot \frac{\theta}{2} \times (\cos \frac{\theta}{2} \sin \frac{\theta}{2} + i \cos^2 \frac{\theta}{2} + i \sin^2 \frac{\theta}{2} - \sin \frac{\theta}{2} \cos \frac{\theta}{2})$$

$$= i \cot \frac{\theta}{2}$$

OR using formulae, $t = \tan \frac{\theta}{2}$

$\cos \theta = \frac{1-t^2}{1+t^2}$

$\sin \theta = \frac{2t}{1+t^2}$

$$\frac{1 + \cos \theta + i \sin \theta}{1 - \cos \theta - i \sin \theta}$$

$$= \frac{1 + \frac{1-t^2}{1+t^2} + \frac{2it}{1+t^2}}{1 - \frac{1-t^2}{1+t^2} - \frac{2it}{1+t^2}} \times \frac{1+t^2}{1+t^2}$$

$$= \frac{1 + \frac{1-t^2}{1+t^2} + \frac{2it}{1+t^2}}{1 - \frac{1-t^2}{1+t^2} - \frac{2it}{1+t^2}} \times \frac{1+t^2}{1+t^2}$$

$$= \frac{1 + \frac{1-t^2}{1+t^2} + \frac{2it}{1+t^2}}{1 - \frac{1-t^2}{1+t^2} - \frac{2it}{1+t^2}} \times \frac{1+t^2}{1+t^2}$$

$$= \frac{2 + 2it}{2t^2 - 2it} \times \frac{(t^2 + it)}{(t^2 + it)}$$

$$= \frac{1 + it}{t^2 - it} \times (t^2 + it)$$

(9) (iii) Hence $\frac{z-1}{z+1}$

$$= \frac{1 + \cos \theta + i \sin \theta}{1 - \cos \theta - i \sin \theta} \times \frac{(1 - \cos \theta - i \sin \theta)}{(1 - \cos \theta - i \sin \theta)}$$

$$\frac{1 + \cos \theta + i \sin \theta}{1 - \cos \theta - i \sin \theta} \times \frac{(1 - \cos \theta - i \sin \theta)}{(1 - \cos \theta - i \sin \theta)}$$

$$= \frac{(2 \cos \theta + 2 i \sin \theta)}{(1 - \cos \theta - i \sin \theta)} \times \frac{(1 - \cos \theta - i \sin \theta)}{(1 - \cos \theta - i \sin \theta)}$$

$$= \frac{2 \cos \theta + 2 i \sin \theta}{2} \times \frac{(1 - \cos \theta - i \sin \theta)}{(1 - \cos \theta - i \sin \theta)}$$

$$= \cos \theta + i \sin \theta$$

(iv) Hence solving $\left(\frac{z-1}{z+1}\right)^4 = -1$

$z = i \cot \left(\frac{\theta}{2}\right)$ where $(\cos \theta + i \sin \theta)^4 = -1$

i.e. $\cos 4\theta + i \sin 4\theta = \cos \pi + i \sin \pi$

Hence $4\theta = \pi$

Wall account $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

as final answer $\left[z = i \cot \frac{\pi}{8}, i \cot \frac{3\pi}{8}, i \cot \frac{5\pi}{8}, i \cot \frac{7\pi}{8} \right]$

As $i \cot \frac{7\pi}{8} = -\cot \frac{\pi}{8}$ & $i \cot \frac{5\pi}{8} = -\cot \frac{3\pi}{8}$

$z = \pm i \cot \frac{\pi}{8}, \pm i \cot \frac{3\pi}{8}$